

Air Motion Detector

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The project described here is of an air motion detector that can warn against strong winds. This is a simple circuit which can be very helpful for people who have no time to check the weather outside. It alerts them to collect their clothes drying in the sun, before strong winds take these away.

Circuit and working

Fig. 1 shows the circuit diagram of the air motion detector alarm. The circuit is built around a piezo sensor element connected across CON2, transistors BC549 (T1), BC547 (T2 through T5) and BC557 (T6), timer NE555 (IC1) in astable mode and other components.

The circuit uses the piezo element used in a buzzer as a sensor. If it senses strong winds, it generates electric pulses. Signals undergo multi-stage amplification by transistors T1, T2 and T3. Amplified signals are rectified and filtered. Obtained DC voltage is again amplified by transistors T4 and T5,

and given to the timer NE555 based oscillator through transistor T6, which is configured as a switch.

Normally, reset pin 4 of IC1 is held at ground level. When the sensor senses strong winds, transistor T6 is switched on, pin 4 of IC1 goes high and the alarm is activated, which produces sounds through the speaker.

Construction and testing

An actual-size, single-side PCB for the air motion detector is shown in Fig. 2 and its component layout in Fig. 3. Enclose the PCB in a suitable small box in such a way that strong winds can push the piezo element. Ensure proper wiring to avoid any mistake.

Test Points

Test point	Details
TP0	0V
TP1	9V provided when switch S1 is closed
TP2	Low (0V) when there is no wind and high (8.8V) when there is strong wind

PARTS LIST

Semiconductors:

- IC1 - NE555 timer
- T1 - BC549 npn transistor
- T2-T5 - BC547 npn transistor
- T6 - BC557 pnp transistor
- D1 - 1N4148 signal diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 1-mega-ohm
- R2 - 4.7-kilo-ohm
- R3 - 220-kilo-ohm
- R4 - 10-kilo-ohm
- R5 - 3.3-kilo-ohm
- R7, R10, R14 - 1-kilo-ohm
- R8 - 10-ohm
- R9 - 100-kilo-ohm
- R11 - 2.7-kilo-ohm
- R12 - 47-ohm
- R13 - 470-ohm
- R15 - 56-kilo-ohm

Capacitors:

- C1 - 1nF ceramic disk
- C2, C4 - 100nF ceramic disk
- C3 - 100 μ F, 25V electrolytic
- C5 - 10 μ F, 25V electrolytic
- C6 - 47nF ceramic disk
- C7 - 1nF ceramic disk
- C8 - 47 μ F, 16V electrolytic

Miscellaneous:

- CON1 - 2-pin connector terminal
- CON2 - 2-pin connector
- S1 - On/off switch
- LS1 - 8-ohm, 0.5W speaker

To ensure proper working of the circuit, check the voltages at various test points given in the table.



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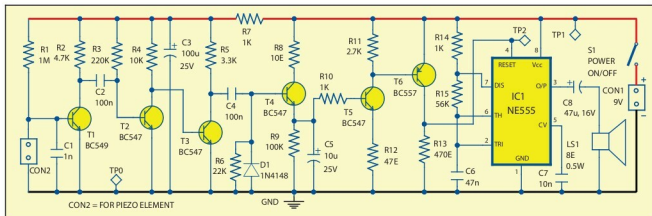


Fig. 1: Circuit diagram of the air motion detector

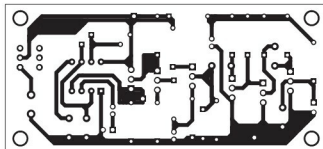


Fig. 2: Actual-size PCB layout of the air motion detector

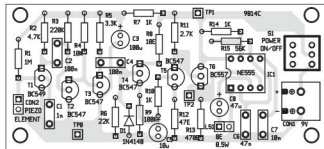


Fig. 3: Component layout of the PCB